

Three phase circuits

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Three phase circuits

Real power, reactive power and apparent power.

- Definitions of real and apparent power.

$$P = |I||V|\cos\phi = |I|^2 R = \frac{|V|^2}{R} = \operatorname{Re}[VI^*]$$

$$Q = |I||V|\sin\phi = |I|^2 X = \frac{|V|^2}{X} = \operatorname{Im}[VI^*]$$

} both conserved

- For inductors $Q > 0$ ($X > 0$), and PF is lagging ($\phi > 0$)

For capacitors $Q < 0$ ($X < 0$), and PF is leading ($\phi < 0$)

- Definition of apparent power.

$$S = |I||V| = |VI^*| = \sqrt{P^2 + Q^2}$$

Star-connected and delta-connected load.

- The phase voltage V_{ph} is the voltage across each phase of the load.

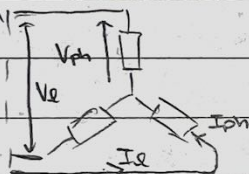
The line voltage V_L is the voltage across a pair of lines.

- The phase current I_{ph} is the current through each phase of the load.

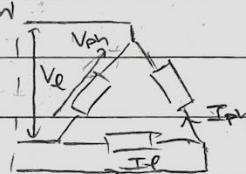
The line current I_L is the current through the line connecting the supply and the load.

- A balanced three-phase load can be connected in star-connection or delta-connection.

star-connection



delta-connection



- For star-connected load,

$$V_L = \sqrt{3} V_{ph}$$

$$I_L = I_{ph}$$

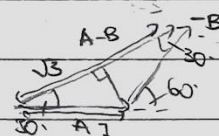
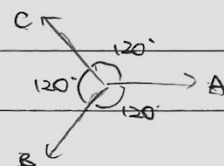
For delta-connected load,

$$V_L = V_{ph}$$

$$I_L = \sqrt{3} I_{ph}$$

- The phase voltage V_{ph} and phase current I_{ph} are related by Ohm's law

$$Z = \frac{V_{ph}}{I_{ph}}$$



- Since the currents form a balanced three-phase set for a balanced three phase load, it doesn't matter whether or not the neutral conductor is connected.

- In a three-phase circuit, the real power P is given by.

$$P = 3 V_{ph} I_{ph} \cos\phi$$

star-connected: $P = 3 \frac{V_L}{\sqrt{3}} I_L \cos\phi = \sqrt{3} V_L I_L \cos\phi$

delta-connected: $P = 3 V_L \frac{I_L}{\sqrt{3}} \cos\phi = \sqrt{3} V_L I_L \cos\phi$

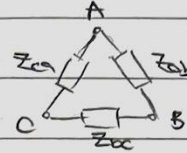
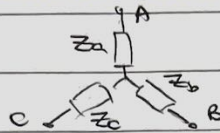
$$P = \sqrt{3} V_L I_L \cos\phi$$

- Similarly for reactive power Q and apparent power S ,

$$Q = 3 V_{ph} I_{ph} \sin\phi = \sqrt{3} V_L I_L \sin\phi$$

$$S = 3 V_{ph} I_{ph} = \sqrt{3} V_L I_L$$

star-delta and delta-star transformation.



- from the OB, $Z_{ab} = \frac{Z_a Z_b + Z_b Z_c + Z_c Z_a}{Z_c}$

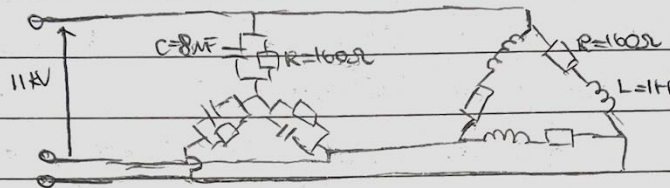
- for balanced loads, $Z_a = Z_b = Z_c = Z_{\text{star}} ; Z_{ab} = Z_{bc} = Z_{ca} = Z_{\text{delta}}$,

$$Z_{\text{delta}} = 3 Z_{\text{star}}$$

$$Z_{\text{star}} = \frac{1}{3} Z_{\text{delta}}$$

Example - three-phase circuit.

- consider the following three-phase circuit.



(i) Find the overall PF, and the line current I_L drawn from the supply.

(ii) The value of the star-connected capacitors C' which, when connected in // w the load, will correct the overall PF to unity.

(i) Method I - use combination of P and Q. (recommended).

star: $P_{\text{star}} = 3 \frac{V_{\text{ph}}^2}{R} = 3 \frac{(11\sqrt{3})^2}{160} = 756.3 \text{ kW}$ $Q_{\text{star}} = -3 \frac{V_{\text{ph}}^2}{X_c} = -\frac{(11\sqrt{3})^2}{\frac{1}{2\pi \cdot 50 \cdot 8\mu}} = -304.1 \text{ kVAR}$

delta: $I_{\text{ph}} = \frac{11\text{K}}{\sqrt{160^2 + (2\pi \cdot 50)^2}} = 31.2 \text{ A}$

$P_{\text{delta}} = 3 I_{\text{ph}}^2 R = 3 \cdot 31.2^2 \cdot 160 = 467.3 \text{ kW}$ $Q_{\text{delta}} = 3 I_{\text{ph}}^2 X_L = 3 \cdot 31.2^2 \cdot (2\pi \cdot 50) = 917.4 \text{ kVAR}$

total: $P_{\text{total}} = P_{\text{star}} + P_{\text{delta}} = 756.3 + 467.3 = 1.22 \text{ MW}$

$Q_{\text{total}} = Q_{\text{star}} + Q_{\text{delta}} = -304.1 + 917.4 = 613 \text{ kVAR}$

$\therefore S_{\text{total}} = \sqrt{P_{\text{total}}^2 + Q_{\text{total}}^2} = \sqrt{1.22^2 + 0.613^2} = 1.37 \text{ MVA}$

$\cos \phi = \frac{P_{\text{total}}}{S_{\text{total}}} = \frac{1.22}{1.37} = 0.894 \text{ (lag)}$

$S_{\text{total}} = \sqrt{3} V_L I_L \rightarrow I_L = \frac{S_{\text{total}}}{\sqrt{3} V_L} = \frac{1.37 \text{ M}}{\sqrt{3} \cdot 11 \text{ K}} = 71.9 \text{ A}$

Method II - use star-delta transformation and use single-phase representation.

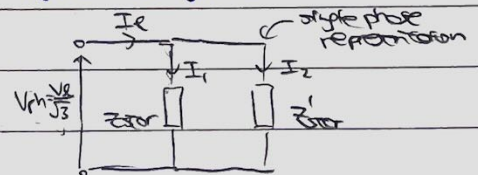
impedance: $Z_{\text{star}} = R // \frac{1}{j\omega C} = 137.7 - j55.18 \Omega$ $Z_{\text{delta}} = R + j\omega L = 160 + j314.2 \Omega$

delta $\rightarrow$ star $Z_{\text{star}'} = \frac{Z_{\text{delta}}}{3} = 53.3 + j104.7 \Omega$

$I_L = \frac{V_{\text{ph}}}{Z_{\text{star}}} + \frac{V_{\text{ph}}}{Z_{\text{star}'}} = 64.23 - j32.19 \text{ A}$

$\phi = -\angle I_L = 26.6^\circ \rightarrow \cos \phi = 0.894 \text{ (lag)}$

$Q_{\text{total}} = \sqrt{3} V_L I_L \sin \phi = 613 \text{ kVAR}$



(ii) $Q_{\text{cap}'} + Q_{\text{total}} = 0$

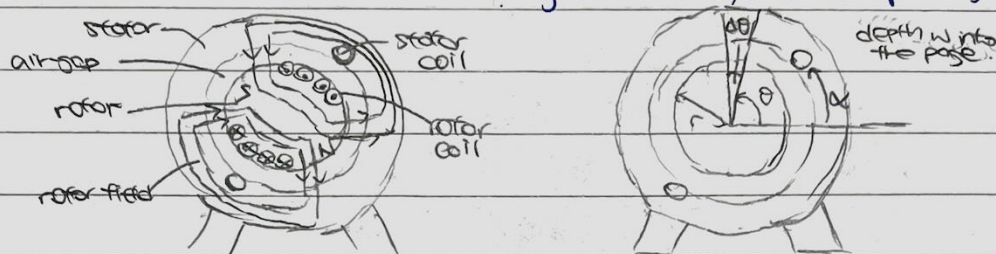
$-3 \frac{V_{\text{ph}}^2}{X_{\text{cap}'}} = 613 \text{ k}$

$C' = 16.1 \mu\text{F}$

Magnetic field of generators.

Magnetic field due to rotor currents.

- A generator consists of a stator and a rotor. Stator is stationary, rotor is supported by bearings and is free to rotate when driven round by the prime mover.
- DC current is fed into the rotor coils via slip-rings and brushes, which set up a magnetic field.



- The magnetic field set up by the DC current varies sinusoidally w/ θ in the air gap.

$$B(\theta) = \hat{B} \cos \theta \quad (\text{rotor stationary})$$

As the rotor is driven round by the prime mover at ω_r ,

$$B(\theta, t) = \hat{B} \cos(\omega_r t - \theta) \quad (\text{rotor rotating})$$

- Consider a stator w/ only one coil. The flux linking an infinitesimal coil area dA is given by

$$d\phi = B dA = B \omega_r r d\theta = \hat{B} \omega_r \cos(\omega_r t - \theta) d\theta$$

Integrating over the whole coil,

$$\phi = \int_{-\pi/2}^{\pi/2} d\phi = \int_{-\pi/2}^{\pi/2} \hat{B} \omega_r \cos(\omega_r t - \theta) d\theta = 2\hat{B} \omega_r \sin(\omega_r t - \alpha)$$

Since the flux changes as the rotor rotates, an emf is induced in the stator coil

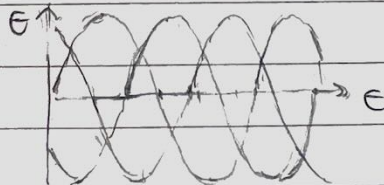
$$e = N \frac{d\phi}{dt} = 2N\hat{B}\omega_r \cos(\omega_r t - \alpha) = \hat{e} \cos(\omega_r t - \alpha)$$

where $\hat{e} = 2N\hat{B}\omega_r$.

Multicore generator.

- Real generators have many coils evenly distributed within the stator. (n coils $\rightarrow 2n$ coil ends)

$n=3$



- Let I_{max} be the max. current per coil (limited by heating).

For a three-phase supply, max. output power is $P_{max} = 3VI_{max}$.

Connecting V_1, V_2, V_3 to give a single-phase supply, max. output power is $P_{max} = 2VI_{max}$.

$\therefore$ efficiency is worse for a single phase device.

no. of phases

no. of wires (L+N)

output efficiency (%)

1

2 (1+1)

66.7

2

3 (2+1)

94.2

3

3 (3+0)

100

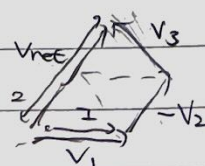
4

4 (4+0)

102

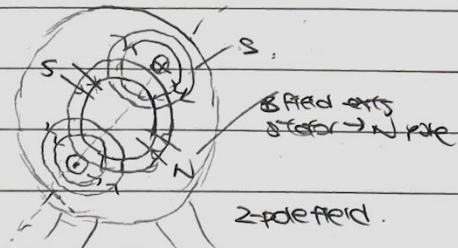
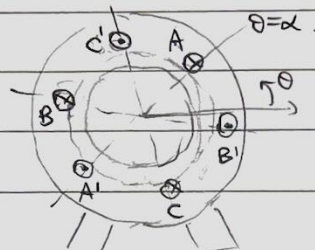
$\rightarrow$ three-phase circuits are the best.

relationship returns



Magnetic field due to stator currents

- Consider the magnetic field due to current in phase A of the stator coil.



Air has low permeability compared to iron $\rightarrow$ magnetic field lines try to min. length across it $\rightarrow$ flux crosses the air gap radially.

- Applying Ampere's law on loop S, we have

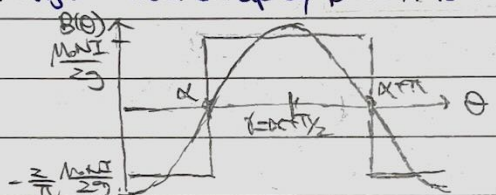
$$\oint \mathbf{H} \cdot d\mathbf{l} = NI$$

$$H_g l_g + H_r l_r = NI$$

$$\frac{B_g}{\mu_0} (2g) + \frac{B_r}{\mu_r \mu_0} l_r = NI$$

$$B = \frac{\mu_r \mu_0 NI}{2g}$$

Therefore we expect the magnetic field set up by phase A to be a square wave w/ amplitude $B = \frac{\mu_r \mu_0 NI}{2g}$.



Taking the first harmonic from the Fourier series expansion of the square wave as an approximation,

$$B(\theta) = \hat{B} \cos(\theta - \gamma)$$

$$\text{where } \hat{B} = \frac{3\mu_r \mu_0 NI}{\pi g}$$

- For three coils spaced 120° apart (A, B, C) and supplied w/ balanced three-phase currents

Phase	γ	$i(\tau)$
A	0	$\hat{I} \cos(\omega\tau)$
B	120°	$\hat{I} \cos(\omega\tau - 120^\circ)$
C	-120°	$\hat{I} \cos(\omega\tau + 120^\circ)$

Superposing the fields they produce,

$$B(\theta, \tau) = B(\theta, \tau)_A + B(\theta, \tau)_B + B(\theta, \tau)_C$$

$$= \frac{3\mu_r \mu_0 \hat{I} N}{\pi g} [\cos\theta \cos\omega\tau + \cos(\theta - 120^\circ) \cos(\omega\tau - 120^\circ) + \cos(\theta + 120^\circ) \cos(\omega\tau + 120^\circ)]$$

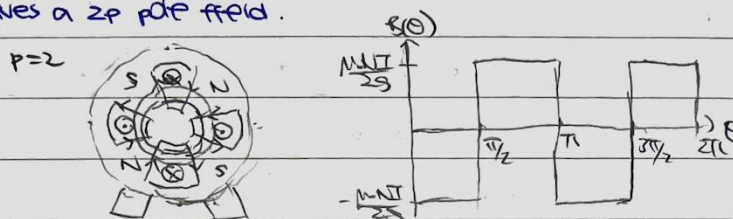
$$= \frac{\mu_r \mu_0 \hat{I} N}{\pi g} [\cos(\omega\tau - \theta) + \cos(\omega\tau + \theta) + \cos(\omega\tau - \theta) + \cos(\omega\tau - \theta - 240^\circ) + \cos(\omega\tau - \theta) + \cos(\omega\tau - \theta + 240^\circ)]$$

$$B(\theta, \tau) = \frac{3\mu_r \mu_0 \hat{I} N}{\pi g} \cos(\omega\tau - \theta)$$

i.e. balanced three-phase currents (exciting each phase sequentially) flowing in a balanced three-phase winding produce a rotating magnetic field.

Multi-pole windings

- p coils per phase gives a $2p$ pole field.



The rotating field is then given by.

$$B(\theta, t) = \hat{B} \cos(\omega t - p\theta)$$

- consider a pt. on the wave and imagine that it moves an angular distance $\Delta\theta$ in time Δt .

$$\hat{B} \cos(\omega t - p\theta) = \hat{B} \cos(\omega(t + \Delta t) - p(\theta + \Delta\theta))$$

$$\therefore \omega t - p\theta = \omega(t + \Delta t) - p(\theta + \Delta\theta)$$

$$\rightarrow \frac{\Delta\theta}{\Delta t} = \frac{\omega}{p} \quad \leftarrow \text{angular frequency of voltage/current}$$

i.e. the speed of rotation of a $2p$ -pole field in the air gap due to stator current is ω/p .
(in general, the speed of rotation of the form above is given by $\frac{\Delta\theta}{\Delta t} = \left| \frac{\text{coeff of } t}{\text{coeff of } \theta} \right|$)

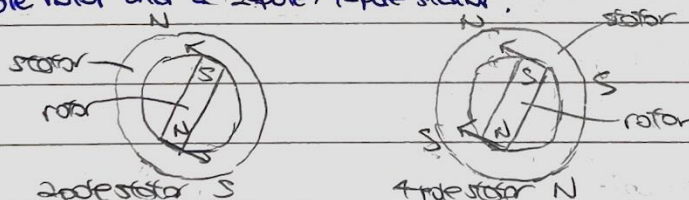
Conditions for steady torque production

- For torque to be produced,

1) the rotor and stator driven fields must have the same no. of poles, $p_r = p_s$

2) the rotor speed must be equal to the speed of the stator-driven field. $\omega_r = \omega/p_s$

①. Consider a 2-pole rotor and a 2-pole/4-pole stator.

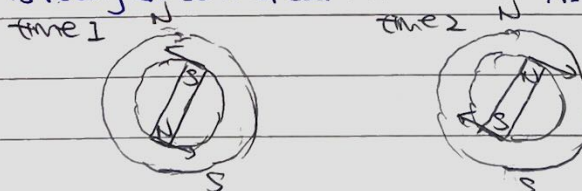


rotor and stator pole no. same $\rightarrow$ net torque acting on the rotor.

rotor and stator pole no. diff. $\rightarrow$ no net torque acting on the rotor.

②. Suppose the stator produces a rotating magnetic field that rotates at a speed ω/p_s .

Assume the rotor is rotating at some speed which is not ω/p_s .



At time 1, torque on rotor is ccw; At time 2, torque on rotor is cw.

$\therefore$ Rotor torque varies w/ time, and has a zero average value.

If rotor speed ω_r is equal to ω/p_s , we have const. torque. This speed is the synchronous speed ω_s .

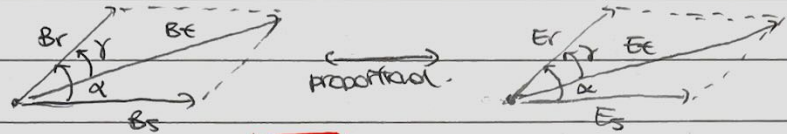
- The torque T produced at synchronous speed is given by

$$T = K \hat{B}_s \hat{B}_r \sin \alpha$$

Equivalent circuit for a generator

Equivalent circuit for a generator.

- The rotor-driven, stator-driven and total air-gap fields (for one of the phases) can be shown on a space phasor diagram.



- We assume an ideal stator winding $\rightarrow E_e = V$

- The stator consists of many coils $\rightarrow$ inductive $\rightarrow E_s \propto B_s \propto I \rightarrow E_s = jX_s I$

where X_s is the synchronous reactance.

- The emf induced by rotor-driven field E_r is known as the excitation voltage $E \rightarrow E_r = E$

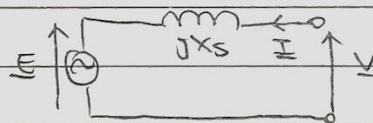
Note that the excitation voltage E is assumed to be prop. to the rotor excitation DC current I_f

$$(E = E_r \propto B_r \propto I_f)$$

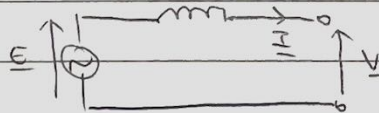
- Therefore we have

$$E_e = E_r + E_s \rightarrow V = E + jX_s I$$

which can be represented by the equivalent circuit.



- Most synchronous machines are operated as generators $\rightarrow$ convenient to note power flowing out of the terminals $\rightarrow$ reverse the direction of current.



$$\therefore E = V + jX_s I$$

stand-alone and parallel operation

- The stator of a synchronous generator can be connected to a three-phase load.

- The frequency of the supply is then fixed by the rotor speed and can be set to any convenient value.

- However, most generators have stators connected to the public electricity supply system, so

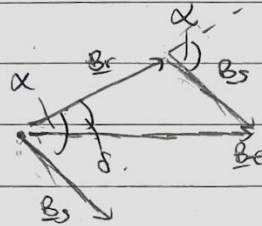
no individual generator can affect its terminal voltage $V \rightarrow V$ has fixed magnitude & frequency.

(This network is known as an infinite bus)

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Variation of torque with load angle

- consider the phase diagram for the magnetic fields.



We define the angle in space between the rotor-driven field and fixed field as the load angle δ .

- Starting from the expression for torque T

$$T = k B_r B_s \sin \alpha.$$

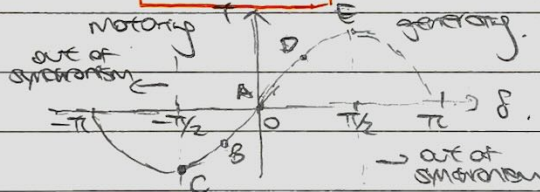
Applying the rule,

$$\frac{\sin(\alpha - \delta)}{B_c} = \frac{\sin \delta}{B_r} \rightarrow B_r \sin \alpha = B_c \sin \delta.$$

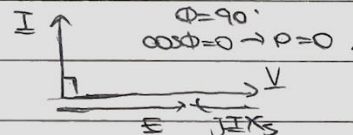
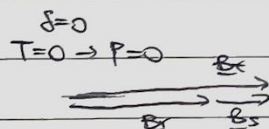
$$\therefore T = k B_r B_c \sin \delta.$$

V is fixed by infinite bus $\rightarrow B_c$ fixed, const. DC excitation $\rightarrow E$ fixed $\rightarrow B_r$ fixed.

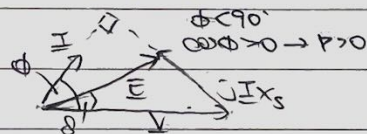
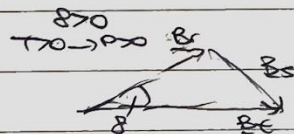
$$T = T_{\max} \sin \delta.$$



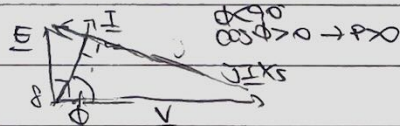
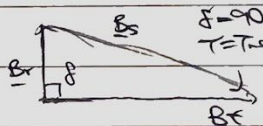
- prime-mover coupled to generator, both rotating at ω_s , but no mech. power transfer (pt. A).



- As prime-mover exerts an accelerating torque on the generator (ω speed fixed to ω_s), the load angle δ increases s.t. generator EM torque = prime-mover torque (pt. D) [generator: mechanical $\rightarrow$ electrical]



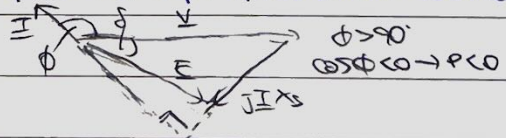
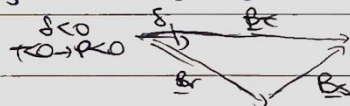
- As prime-mover torque increased, the load angle δ increases until $\delta = 90^\circ$ to reach max. EM torque (pt. E).



(any increase in prime-mover torque causes synchronous machine to lose synchronism).

- Suppose a mechanical load is coupled to the synchronous machine $\rightarrow$ it applies a braking torque ($T < 0$).

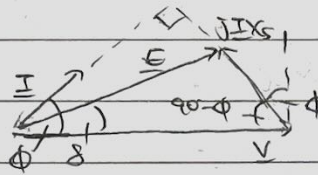
the load angle δ decreases s.t. generator EM torque = prime-mover torque (pt. B) [motor: electrical $\rightarrow$ mechanical]



As the mechanical load increases and the load angle δ decreases until $\delta = -90^\circ$, max. output power is reached. (any increase in mechanical load causes synchronous machine to lose synchronism)

Power and torque in terms electrical quantities

- consider the phasor diagram for the voltage.



the output electrical power of the generator is $P_{out} = 3VI\cos\phi$

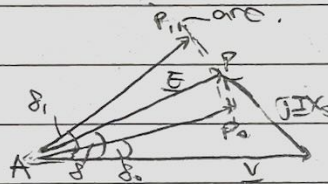
the input mechanical power of the generator is $P_{in} = T\omega_s$

Neglecting only power losses, $P_{in} = P_{out} \rightarrow 3VI\cos\phi = T\omega_s \rightarrow T = \frac{3VI\cos\phi}{\omega_s}$

From the phasor diagram, $E\sin\delta = X_s I \cos\phi \rightarrow I \cos\phi = \frac{E\sin\delta}{X_s} \rightarrow T = \frac{3VE}{\omega_s X_s} \sin\delta$

Control of real power

- since V is fixed (infinite bus), and E is fixed (fixed excitation DC voltage), pt. P on the phasor diagram is constrained to move in an arc about A.



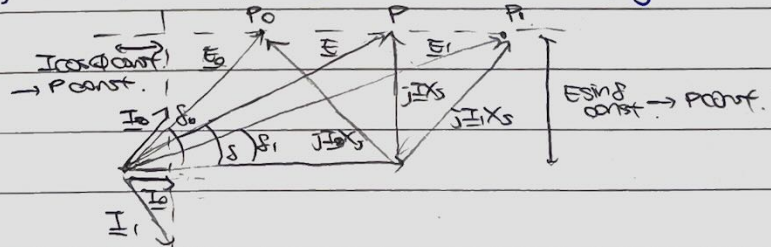
- As the power changes from P_0 to P_1 ,

$$\frac{P_1}{P_0} = \frac{T_1 \omega_s}{T_0 \omega_s} = \frac{3VE\sin\delta_1/X_s}{3VE\sin\delta_0/X_s} = \frac{\sin\delta_1}{\sin\delta_0}$$

i.e. controlling load angle δ controls real power P . ($\uparrow\delta \rightarrow \uparrow P$).

Control of reactive power

- since $P = T\omega_s = \frac{3VE\sin\delta}{X_s}$, to keep P const. $\rightarrow E\sin\delta = \text{const} \rightarrow$ vertical proj. of P const.

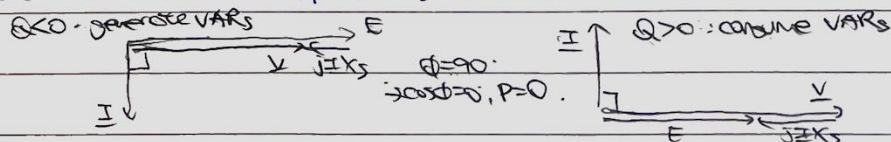


- Increasing $|E| \rightarrow$ pt. P moves to pt. $P_1 \rightarrow$ current lags voltage $\rightarrow$ machine supplies VARs, overexcited.

Decreasing $|E| \rightarrow$ pt. P moves to pt. $P_0 \rightarrow$ current leads voltage $\rightarrow$ machine absorbs VARs, underexcited.

i.e. controlling excitation voltage $|E|$ controls reactive power Q ($\uparrow|E| \rightarrow \downarrow Q$)

- A synchronous compensator is a synchronous machine w/ $P=0$ and $Q \neq 0$. ($P=0 \rightarrow \delta=0$).



- when $|E| > |V|$ (over-excited), the machine supplies reactive power

when $|E| < |V|$ (under-excited), the machine absorbs reactive power.

Operation of a generator

construction of a generator.

- stator's experience changing magnetic field $\rightarrow$ laminated to min eddy current losses.
- Rotor either fitted w/ slip-rings + brushes or a AC generator + rectifier to supply the field current I_f .
- Two types of rotor constructions are used:
 - $\hookrightarrow$ cylindrical rotor - high speed machines.
 - $\hookrightarrow$ salient-pole rotor - low speed machines.

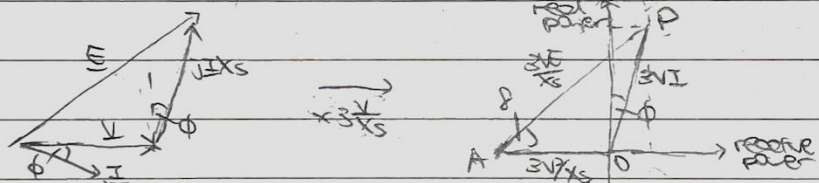
Rating of a generator.

- The generator current I_{ph} is limited by the heating of stator coils due to resistance (cu losses).
- The generator voltage V_{ph} is limited by the flux (Fe losses).

$\therefore$ Generators have a VA rating $S_{max} = 3V_{ph}I_{ph}$

operating chart:

- Scale the phasor diagram for the voltages by $\frac{3V}{X_s}$ to get the operating chart

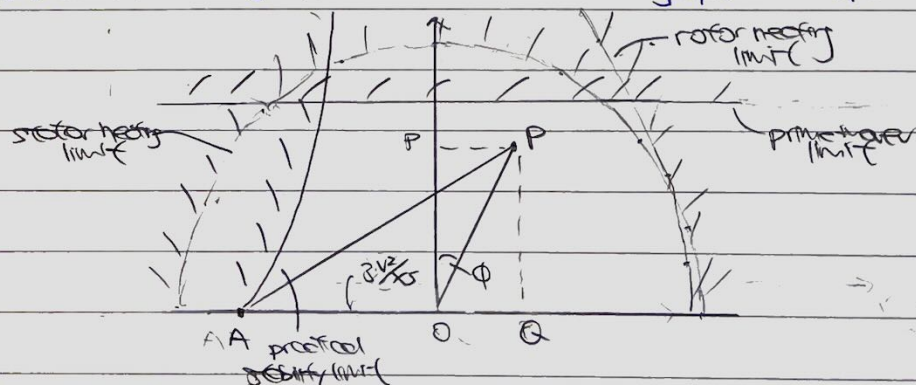


The vertical projection of OP is $3VI\cos\phi = P$

The horizontal projection of OP is $3VI\sin\phi = Q$

- There are restrictions to the generator's operation chart

- $\hookrightarrow$ 1) Length $AO = \frac{3V}{X_s}$ is fixed for operation from the infinite bus (V_{max} , X_s fixed).
- $\hookrightarrow$ 2) Real power limited by prime mover $\rightarrow$ horizontal line $P = P_{max}$ [prime-mover limit]
- $\hookrightarrow$ 3) Length $OP = 3VI = S \leq S_{max}$, where S_{max} is the rated VA of the generator $\rightarrow$ draw a circle of radius S_{max} , centred O [stator-heating limit].
- $\hookrightarrow$ 4) Length $AP = \frac{3VE}{X_s} \leq \frac{3VE_{max}}{X_s}$, where E_{max} is the max. excitation emf $\rightarrow$ draw a circle of radius $\frac{3VE_{max}}{X_s}$, centred A [rotor-heating limit].
- $\hookrightarrow$ 5) Load angle $\delta \ll 90^\circ$. Load angle δ close to 90° would risk losing synchronism [practical stability limit]



Power transmission

Per-unit system

- The per unit value is calculated by

$$\text{per unit value} = \frac{\text{actual value of quantity}}{\text{base value of quantity}}$$

- By convention, we choose base values for VA and voltage, then we derive base values for current and impedance:

→ choose base voltage V_b and base VA S_b

→ derive base current I_b and base impedance Z_b .

$$S = \sqrt{3} V_L I_L \rightarrow S_b = \sqrt{3} V_b I_b \rightarrow I_b = \frac{S_b}{\sqrt{3} V_b}$$

$$Z = \frac{V_{ph}}{I_{ph}} = \frac{V_L/\sqrt{3}}{S/\sqrt{3} V_L} = \frac{V_L^2}{S} \rightarrow Z_b = \frac{V_b^2}{S_b}$$

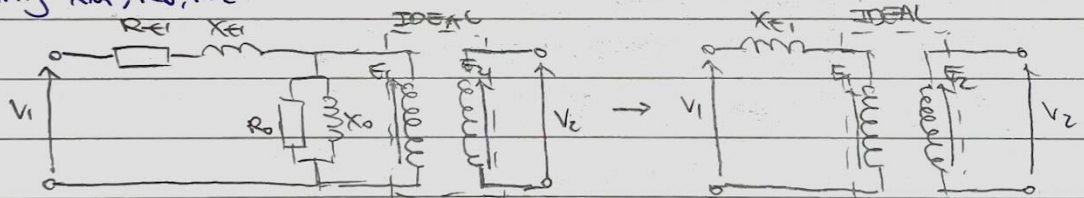
- Using these base values,

$$S = \sqrt{3} V_L I_L = \sqrt{3} V_b V_{ph} I_b I_{ph} = \sqrt{3} V_b V_{ph} \frac{S_b}{\sqrt{3} V_b} \frac{I_{ph}}{I_b} = S_b V_{ph} I_{ph}$$

$$\therefore S_{pu} = \frac{S}{S_b} = V_{ph} I_{pu}$$

Representation of transformers.

- In the context for power transmission, the equivalent circuit for transformers can be simplified by ignoring X_m , R_0 , R_E .



Take transformer rated VA as common S_b , and primary rated line voltage as V_{b1} .

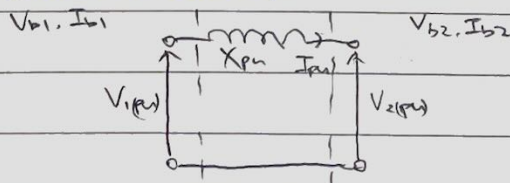
Then choose V_{b2} using the turns ratio: $\frac{V_{b2}}{V_{b1}} = \frac{N_2}{N_1} = \frac{E_2}{E_1}$.

$$\therefore \frac{E_1}{V_{b1}} = \frac{E_2}{V_{b2}} \rightarrow E_{2(pu)} = E_{1(pu)}$$

Also, considering the ratio of currents at primary and secondary, $\frac{I_1}{I_2} = \frac{N_2}{N_1} = \frac{V_{b2}}{V_{b1}} = \frac{I_{b1}}{I_{b2}}$

$$\therefore \frac{I_1}{I_{b1}} = \frac{I_2}{I_{b2}} \rightarrow I_{2(pu)} = I_{1(pu)}$$

The transformer equivalent circuit then becomes.



i.e. provided base voltages throughout the system are chosen in the ratio of transformer turns, the pu system gets rid of all the ideal transformers.

change of base.

- Sometimes, it is necessary to express a pu impedance to a diff. base:

say, Base 1: V_{b1}, S_{b1} ; Base 2: V_{b2}, S_{b2}

$$Z_{1(pu)} = \frac{Z(\Omega)}{Z_{b1}} = \frac{Z(\Omega) S_{b1}}{V_{b1}^2} \quad Z_{2(pu)} = \frac{Z(\Omega)}{Z_{b2}} = \frac{Z(\Omega) S_{b2}}{V_{b2}^2}$$

Impedance in Ω must not depend on the base.

$$\therefore Z(\Omega) = Z_{1(pu)} \frac{V_{b1}^2}{S_{b1}} = Z_{2(pu)} \frac{V_{b2}^2}{S_{b2}}$$

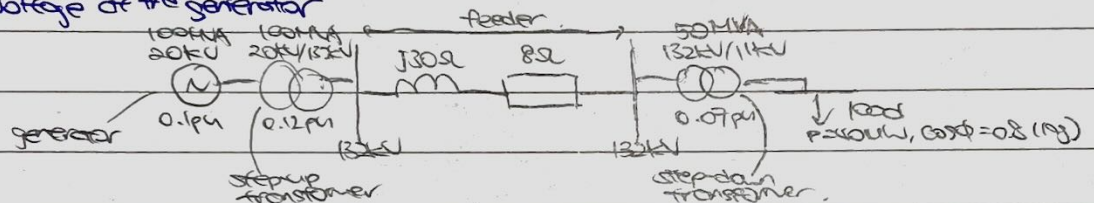
$$Z_{2(pu)} = Z_{1(pu)} \cdot \frac{S_{b2}}{S_{b1}} \cdot \frac{V_{b1}^2}{V_{b2}^2}$$

Often, change of base only due to change of S_b , i.e. $V_{b1} = V_{b2}$, so

$$Z_{2(pu)} = Z_{1(pu)} \cdot \frac{S_{b2}}{S_{b1}}$$

Example - transmission line

- Find the real and reactive power supplied by the generator, power dissipated in the feeder and the voltage of the generator



choose $S_b = 100 \text{ MVA}$, $V_b = 20 \text{ kV}, 132 \text{ kV}, 11 \text{ kV}$.

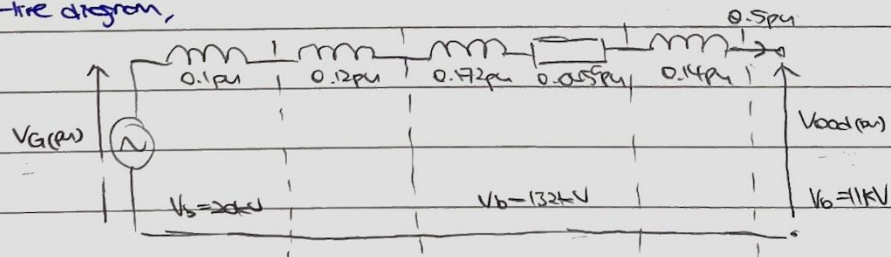
change of base on 50MVA transformer: $Z_{pu(100)} = Z_{pu(50)} \cdot \frac{100}{50} = 0.14 \text{ pu}$.

convert feeder impedance to pu: $Z_b = \frac{V_b^2}{S_b} = \frac{(132 \text{ kV})^2}{100 \text{ MVA}} = 174 \Omega$; $Z_{pu} = \frac{Z(\Omega)}{Z_b} = 0.0459 + j0.172$

At the load, $S = \frac{P}{\cos\phi} = \frac{40 \text{ MW}}{0.8} = 50 \text{ MVA}$; $S_{pu} = \frac{S}{S_b} = \frac{50 \text{ MVA}}{100 \text{ MVA}} = 0.5$

$$S_{pu} = V_{pu} I_{pu} \rightarrow I_{pu} = 0.5$$

Drawing the single-line diagram,



The total series pu reactance $X_{T(pu)} = 0.1 + 0.12 + 0.172 + 0.0459 + 0.14 = 0.532$.

conv. of P: $P_{f(pu)} = I_{pu}^2 R_{f(pu)} = 0.5^2 \cdot 0.0459 = 0.0115$; $P_f = P_{f(pu)} \cdot S_b = 1.15 \text{ MW}$,

$P_{G(pu)} = P_{f(pu)} + P_{load(pu)} = 0.0115 + \frac{40 \text{ MW}}{100 \text{ MVA}} = 0.4115$; $P_G = P_{G(pu)} \cdot S_b = 41.15 \text{ MW}$.

conv. of Q: $Q_{G(pu)} = I_{pu}^2 X_{T(pu)} + Q_{load(pu)} = 0.5^2 \cdot 0.532 + \frac{30 \text{ MVAR}}{100 \text{ MVA}} = 0.433$;

$Q_G = Q_{G(pu)} \cdot S_b = 43.3 \text{ MVAR}$.

$$S_G(pu) = V_G(pu) I_G(pu)$$

$$\sqrt{P_{G(pu)}^2 + Q_{G(pu)}^2} = V_G(pu) I_G(pu) \rightarrow V_G(pu) = 1.195$$

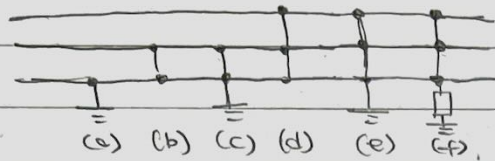
$$\therefore V_G = V_G(pu) \cdot V_b = 23.9 \text{ kV}$$

Power system faults.

Power system faults.

- Power system faults are caused by lightning, weather (gales/snow), insulation error, people.

- Types of faults include.



we will only consider three-phase symmetrical fault to earth, i.e. type (e).

Protection.

- The protection of power systems is achieved using circuit breakers and isolators.

↳ circuit breakers: opens on load

↳ isolators: only opens offload.

- The sequence of events following a fault: fault occurs → sensors detect → circuit breakers open → recloses → reopens if fault not cleared → isolators open.

- The breaker opens by moving metal contacts apart → causes arc → current falls to zero → remove ionisation products by blast of fluid (air, oil, SF₆) to prevent arc restriking.

- The isolator in series w/ circuit-breaker opens when current has been interrupted.

- A protection system should have the following qualities:

↳ Discrimination: only isolates the faulty part of system

↳ Stability: breaker stays closed when fault is outside the protected zone.

↳ Sensitivity: ability to set current level at which break opens

↳ Speed: breaker opens quickly enough to prevent damage to generators, transformers etc.

↳ Reliability: breakers should not open when there is no fault (15% of faults are short-term)

↳ Back up: separate system in case primary protection fails.

Fault analysis.

- We can use fault analysers to determine fault current levels I_f → specify breaker VA ratings,

defined as $S_F = \sqrt{3} V_{\text{line}} I_f$ ($S_{F(p.u)} = V_{\text{line}}(p.u) I_{f(p.u)}$)

- In fault analysis, we assume the following:

↳ All generators can be modelled as $E_{pu} = 1 \angle 0^\circ$ emf in series w/ synchronous reactance

↳ Transmission lines and transformers can be modelled as series inductance reactances ($R=0$).

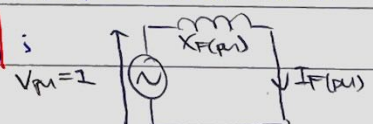
↳ The effect of loads connected to the network can be ignored.

- We reduce the network to a Thevenin equivalent, and find $X_{F(p.u)}$, and hence $I_{f(p.u)}$.

$$I_{f(p.u)} = \frac{V_{pu}}{X_{F(p.u)}} = \frac{1}{X_{F(p.u)}};$$

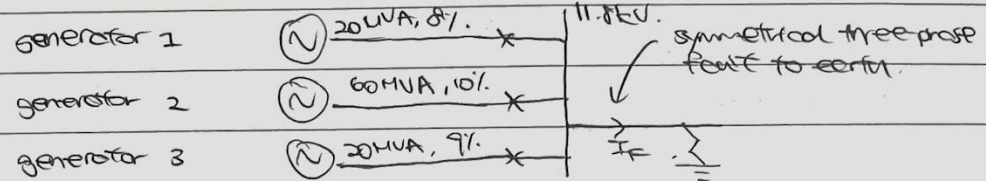
$$S_{F(p.u)} = V_{pu} I_{f(p.u)} = \frac{1}{X_{F(p.u)}};$$

$$\therefore I_f = I_{f(p.u)} \cdot I_b = \frac{I_b}{X_{F(p.u)}}; S_F = S_{F(p.u)} \cdot S_b = \frac{S_b}{X_{F(p.u)}}$$



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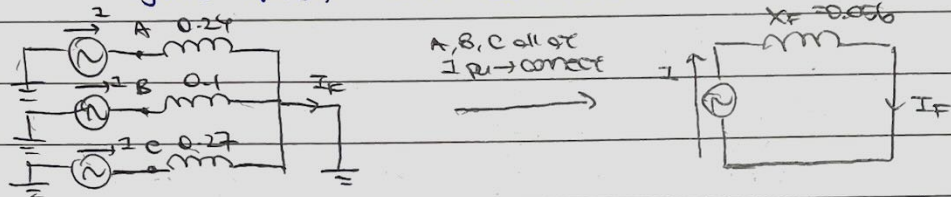
-Example: Find the fault current and fault VA and VA rating of generator circuit breakers.



Choose $S_b = 60 \text{ MVA}$, $V_b = 11.8 \text{ kV}$

change of base: gen 1 $X_{pu}(w) = 0.08 \cdot \frac{60}{20} = 0.24$

gen 3 $X_{pu}(w) = 0.09 \cdot \frac{60}{20} = 0.27$



$$I_{F(pu)} = \frac{1}{X_{F(pu)}} = \frac{1}{0.56} = 1.79 \quad ; \quad I_b = \frac{S_b}{\sqrt{3} V_b} = 2.9 \text{ kA} \quad \text{req. VA rating for 1 style breaker}$$

$$\therefore I_F = I_{F(pu)} \cdot I_b = 52.5 \text{ kA} \quad ; \quad S_F = S_{F(pu)} \cdot S_b = \frac{S_b}{X_{F(pu)}} = 1.07 \text{ GVA}$$

$$\text{For each generator, } S_{G(pu)} = V_{pu} I_{G(pu)} = \frac{1}{X_{G(pu)}} \rightarrow S_G = S_{G(pu)} \cdot S_b = \frac{S_b}{X_{G(pu)}}$$

$$\therefore \text{gen 1 } S_G = 250 \text{ MVA} \quad , \quad \text{gen 2 } S_G = 600 \text{ MVA} \quad , \quad \text{gen 3 } S_G = 222 \text{ MVA}$$

Induction motors

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Operation of induction motors

Principles of operation. of induction motors

- An induction motor has a stator wound w/ a balanced three-phase winding, which is excited by a balanced three-phase voltage supply $\rightarrow$ a rotating magnetic field is set up in the air gap.

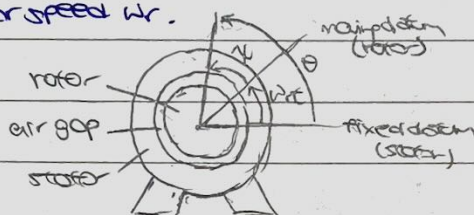
$$B_s(\theta, t) = \hat{B}_s \cos(\omega_s t - p\theta)$$

[speed of rotation of stator-driven field ω_s]

and the field rotates at synchronous speed $\omega_s = \frac{\omega}{p}$ $\leftarrow$ angular frequency of voltage supply.

- There is also a rotor wound w/ a balanced three-phase winding, and the terminals short-circuited. The rotor rotates at angular speed ω_r .

[speed of rotation of rotor ω_r]



- Viewed from the rotor, the stator-driven field appears the same as when viewed from the stator, except rotating at speed $\omega_{sr} = \omega_s - \omega_r$.

- Defining γ as an angular coordinate fixed to the rotor, the stator-driven field is given by.

$$B_s(\gamma, t) = \hat{B}_s \cos((\omega - p\omega_r)t - p\gamma)$$

- Defining slip s as

$$s = \frac{\omega_s - \omega_r}{\omega_s} \rightarrow \omega_r = (1-s)\omega_s$$

substituting into the expression for the stator-driven field, we have

$$B_s(\gamma, t) = \hat{B}_s \cos(s\omega_s t - p\gamma)$$

and the field rotates at $\omega_s - \omega_r = s\omega_s = \frac{s\omega}{p}$ wrt the rotor.

- The stator-driven field changes wrt the rotor $\rightarrow$ balanced three-phase e.m.f. of frequency

$\omega_s - \omega_r = s\omega_s = \frac{s\omega}{p}$ induced in the rotor winding $\rightarrow$ balanced three-phase currents of frequency

$\omega_s - \omega_r = s\omega_s = \frac{s\omega}{p}$ flow $\rightarrow$ set up a rotor-driven magnetic field.

$$B_r = \hat{B}_r \cos(s\omega_s t - p\gamma - \alpha)$$

and the rotor-driven field rotates at speed $\omega_s - \omega_r = s\omega_s = \frac{s\omega}{p}$ wrt the rotor.

- The speed of rotation of the rotor-driven field in the stationary reference frame is.

$$\omega_{rq} = \omega_r + (\omega_s - \omega_r) = \omega_s$$

[speed of rotation of rotor-driven field ω_r]

- * stator and rotor-driven fields have the same no. of poles, and rotate at the same speed (ω_s).

$\rightarrow$ conditions for steady torque to be produced.

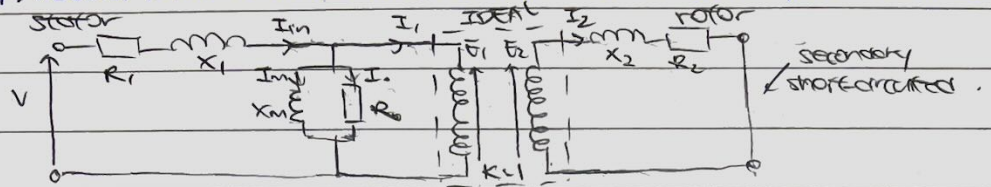
two fields interact to produce torque

- If the rotor rotates at synchronous speed (i.e. $\omega_r = \omega_s$), the stator-driven field is stationary in the rotor's reference frame $\rightarrow$ no e.m.f. induced in rotor winding $\rightarrow$ no current $\rightarrow$ no rotor-driven field $\rightarrow$ no torque.

- Torque is produced at all other rotor speeds (i.e. $\omega_r \neq \omega_s$) $\rightarrow$ asynchronous motor.

Equivalent circuit for induction motors

- When stationary, induction motor is similar to a transformer w/ the secondary short-circuited.



R_1 : stator winding resistance

R_2 : rotor winding resistance

X_1 : stator leakage reactance

X_2 : rotor leakage reactance

R_0 : iron loss resistance

X_m : magnetising reactance

define k as the effective turns ratio, $K = \frac{N_{\text{stator}}}{N_{\text{rotor}}} = \frac{E_1}{E_2} = \frac{I_2}{I_1}$

- The induced e.m.f per phase in the stationary rotor is

$$E_2 = I_2 (R_2 + jX_2)$$

- Now, allow the rotor to rotate w/ slip s ($\omega_r = (1-s)\omega_s$). speed of rotation of stator-driven field rel. to the rotor drops from ω_s to $s\omega_s$.

$$E_2 \propto \omega \text{ so } E_2 \rightarrow sE_2$$

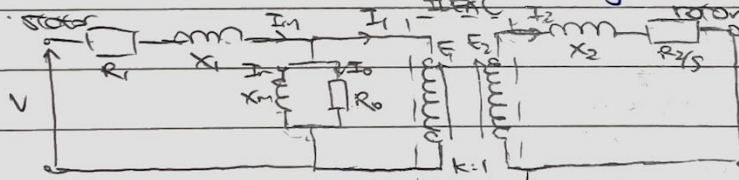
$$X_2 \propto \omega \text{ so } X_2 \rightarrow sX_2$$

$$\therefore sE_2 = I_2 (R_2 + jsX_2)$$

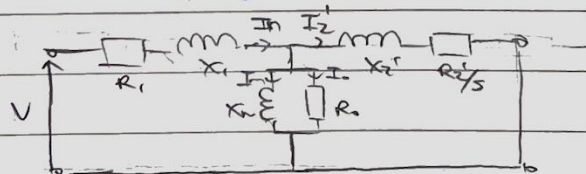
Dividing by s , we get the induced e.m.f per phase in the rotating rotor.

$$E_2 = I_2 \left(\frac{R_2}{s} + jX_2 \right)$$

- The equivalent circuit for the induction motor when rotating w/ slip s is

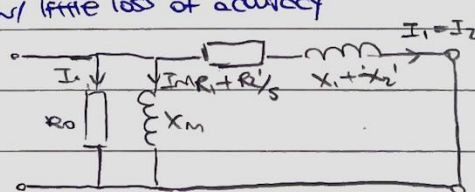


referring the rotor parameters to the stator (secondary referred to primary)..



$$\text{where } X_2' = K^2 X_2 ; R_2' = K^2 R_2$$

& sometimes, we can assume the voltage drop across $R_1 + jX_1$ is small $\rightarrow$ swap places w/ the magnetising branch R_0, X_m w/ little loss of accuracy



(useful for calculating I_2' for the expression for torque, $T = \frac{3}{\omega_s} I_2'^2 \frac{R_2'}{s}$)

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Determination of equivalent circuit parameters

① No-load test (counterpart to open-circuit test)

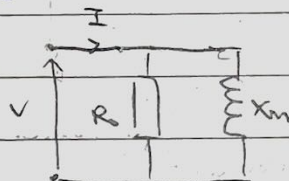
- In the no-load test, the motor is supplied the rated voltage, but with no mechanical load.

$$N_r \rightarrow N_s \quad \text{so} \quad s \rightarrow 0 \quad \text{so} \quad R_2'/s \rightarrow \infty.$$

Since the impedance of the // branch containing R_0 and X_m is far larger than the series components

R_1 and $X_1 \rightarrow R_0$ and X_1 may be ignored.

The equivalent circuit is therefore



Real power only consumed in R_0 : $P = 3 \frac{V^2}{R_0} \rightarrow R_0 = 3 \frac{V^2}{P}$

Reactive power only consumed in X_m : $Q = 3 \frac{V^2}{X_m} \rightarrow X_m = 3 \frac{V^2}{Q}$

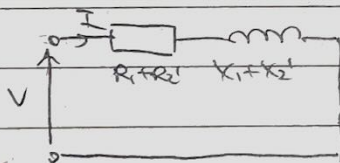
② Locked-rotor test (counterpart to short-circuit test).

- In the locked-rotor test, the rotor is held stationary.

$$s = 1 \quad \text{so} \quad R_2'/s \text{ is small.}$$

so the parallel components R_0 and X_m may be neglected in comparison to the series components R_2 and X_2 .

The equivalent circuit is therefore.



Real power only consumed in R_1+R_2' : $P = 3 I^2 (R_1+R_2') \rightarrow R_1+R_2' = 3 \frac{P}{I^2}$

Reactive power only consumed in X_1+X_2' : $Q = 3 I^2 (X_1+X_2') \rightarrow X_1+X_2' = 3 \frac{Q}{I^2}$

+ R_1 is usually found with an ohmmeter of the stator; $X_1 = X_2'$ is usually given.

Construction of induction motors.

- The stator carries a three-phase winding and is laminated (min. eddy current losses).

- The rotor winding has the same no. of poles as the stator and produce a rotating magnetic field.

It is also laminated (min. eddy current losses). Two options for the rotor are:

↳ wound rotor

↳ cage rotor.

Power and torque of induction motors

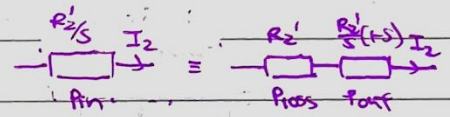
Power and torque in induction motors.

- Input power of induction motor is given by

$$P_{in} = 3I_1^2 R_1 + 3I_0^2 R_0 + 3I_2'^2 \frac{R_2'}{s}$$

- The power losses in the induction motor is given by

$$P_{loss} = \underbrace{3I_1^2 R_1}_{\text{stator winding loss}} + \underbrace{3I_0^2 R_0}_{\text{iron loss}} + \underbrace{3I_2'^2 \frac{R_2'}{s}}_{\text{rotor winding loss}}$$



- Applying conservation of power,

$$P_{in} = P_{out} + P_{loss}$$

$$\therefore P_{out} = 3I_2'^2 \frac{R_2'}{s} (1-s)$$

- Mechanical power is given by $T\omega_r = T\omega_s(1-s)$, so

$$3I_2'^2 \frac{R_2'}{s} (1-s) = T\omega_s(1-s)$$

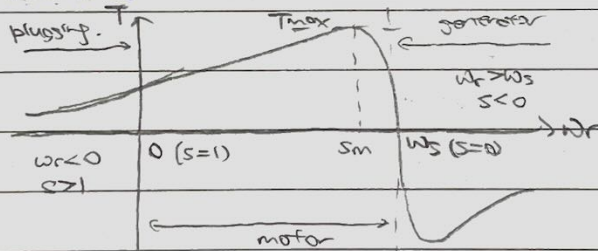
$$T = \frac{3}{\omega_s} I_2'^2 \frac{R_2'}{s}$$

where T is the gross torque. There is also a mechanical loss torque (e.g. friction, wind by cooling fan)

$$T_{net} = T - T_{loss}$$

Variation of torque with speed (slip)

- The variation of torque w/ speed is as follows:

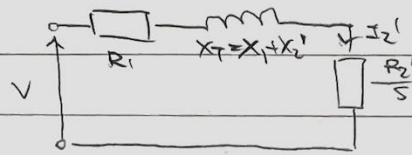


The max torque T_{max} is achieved at slip s_m .

- For $0 < s < 1$, electrical $\rightarrow$ mechanical : motor
- $s < 0$, mechanical $\rightarrow$ electrical : generator
- $s > 1$, electrical $\rightarrow$ losses : plugging.

Maximum torque

- Moving the magnetizing branch R_0, X_m to the terminals, we have.



Applying max. power transfer theorem, we get max. power dissipated (and thus torque) in R_2'/s when

$$\frac{R_2'}{s_m} = Z_s \rightarrow \boxed{s_m = \frac{R_2'}{Z_s}}$$

where $Z_s = \sqrt{R_1^2 + X_T^2}$ and $X_T = X_1 + X_2'$.

- Apply Ohm's law to find the current I_{2m}' .

$$I_{2m}' = \frac{V}{\sqrt{(R_1 + \frac{R_2'}{s_m})^2 + X_T^2}} = \frac{V}{\sqrt{(R_1 + Z_s)^2 + X_T^2}}$$

$$\therefore I_{2m}'^2 = \frac{V^2}{(R_1 + Z_s)^2 + X_T^2} = \frac{V^2}{Z_s^2 + 2Z_s R_1 + R_1^2 + X_T^2} = \frac{V^2}{2Z_s(Z_s + R_1)}$$

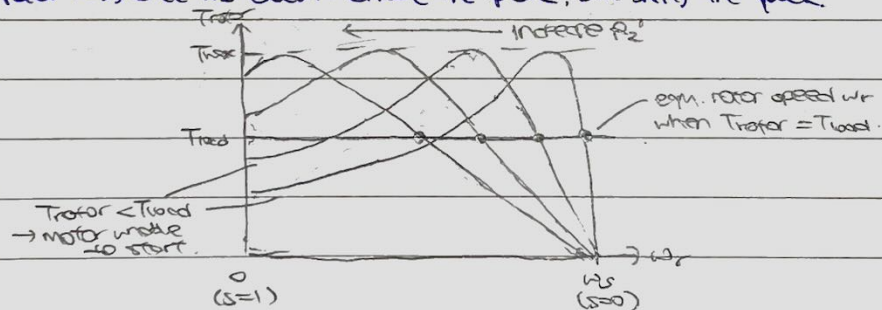
$$\rightarrow T_{max} = \frac{3}{\omega_s} I_{2m}'^2 \frac{R_2'}{s_m} = \frac{3}{\omega_s} \frac{V^2}{2Z_s(Z_s + R_1)} \cdot Z_s$$

$$\boxed{T_{max} = \frac{3V^2}{2\omega_s(Z_s + R_1)}}$$

Rotor resistance and torque/speed control

- From above, we find that T_{max} is independent of R_2' and $s_m \propto R_2'$.

→ adding rotor resistance R_2' doesn't affect the peak, but shifts the peak.



- The load only accelerates when $T_{max} > T_{load} \rightarrow \omega_r$ increases until $T_{motor} = T_{load}$

- shifting the peak (to the left) by increasing rotor resistance R_2' can increase the torque when starting ($\omega_r = 0$), and thus decrease the starting current.

* For max. starting torque, $s_m = 1 \rightarrow \frac{R_2'}{Z_s} = 1 \rightarrow R_2' = Z_s$.

- For wound-rotor machines, extra rotor resistance can be connected via slip rings.

For cage-rotor machines, the skin effect is exploited.